Shared Structure Through Alignment: Testing Platonic Scaling in Cross-Survey Astronomy

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Abstract

The Platonic Representation Hypothesis predicts increasing representational convergence with model scale, but the observed trend depends on how cross-system correspondence is measured. We study matched Legacy Survey and HSC astronomical embeddings across five model families and sixteen model-size rungs. A supervised dense linear alignment substantially increases held-out neighbourhood correspondence and steepens its dependence on model size relative to native cosine geometry: the slope increase is positive in all five families (T=0.00721), survives an object-level bootstrap (95% CI $[0.0034, 0.0111]$), and exceeds a synchronized correspondence-shuffle refit null (p=$1/201 \approx 0.005$). The effect is not explained by rotation or change of basis; the learned maps remain strongly anisotropic with size, with most useful transfer concentrated in a small leading singular subspace. PCA bottlenecks further improve alignment level but do not clearly strengthen scaling, while sparse SAE and BSF representations do not outperform the matched dense-alignment baseline. These results suggest that model scaling increases linearly recoverable cross-survey structure without driving the representations toward a common isometric geometry.

1 Introduction

Representations from independently trained models often become more aligned as capability and scale increase [Huh et al., 2024]. The Platonic Representation Hypothesis (PRH) interprets this as convergence toward a shared statistical model of reality. In astronomy, matched survey embeddings give a comparatively clean test: prior work reports increasing mutual k-nearest-neighbour (mKNN) agreement with model capacity on Legacy Survey [Dey et al., 2019]$\leftrightarrow$HSC [Miyazaki et al., 2018] ladders [UniverseTBD et al., 2025].

A strong reading of Platonic convergence asks whether Legacy and HSC embeddings X_L and X_H become directly geometrically similar. A weaker, but often more operational, reading asks whether they become increasingly equivalent under a restricted transformation class $T(X_L) \approx X_H$. We organize allowed maps as $\mathcal{T}_{\text{id}} \subset \mathcal{T}_{\text{similarity}} \subset \mathcal{T}_{\text{linear}}$: identity/native geometry; rotation/reflection plus uniform scaling (similarity/isometric class), which preserves all pairwise cosines and therefore cosine kNN structure; and general linear maps, which can apply direction-dependent stretch. PCA is treated as a *bottleneck before* linear alignment rather than as another nested geometric class. We distinguish three measurements on the same held-out objects: *native correspondence* M_{Dense} (cosine geometry in the original coordinates); *linearly recoverable correspondence* $M_{\text{Dense+Ridge}}$ (after a supervised global linear map fit on paired training objects); and *bottleneck-assisted recoverability* $M_{\text{PCA+Ridge}}$ (independent survey PCAs before the same map). We also evaluate sparse SAE [Gao et al., 2025]/BSF [Fel et al., 2026] representations with matched Ridge supervision and—as a complementary construction—an unpaired nonlinear shared bottleneck [Jha et al., 2025].

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The central empirical question is which notion of cross-survey equivalence exhibits model-size scaling. We find that supervised linear recoverability shows substantially stronger size dependence than native cosine geometry, without evidence that the required maps approach rotations or other similarity transforms.

2 Setup

We use the official UniverseTBD Legacy↔HSC matched embeddings [UniverseTBD et al., 2025] (n=16,384 objects) across five families—AstroPTv2 [Smith et al., 2024], ConvNeXtv2 [Woo et al., 2023], DINOv2 [Oquab et al., 2024], ViT [Dosovitskiy et al., 2021], and I-JEPA [Assran et al., 2023]—sixteen size rungs in total (Appendix A). I-JEPA has only two rungs; its slopes are two-point.

For representations A and B,

$$M_k = \frac{1}{n} \sum_i \frac{|\mathcal{N}_k^A(i) \cap \mathcal{N}_k^B(i)|}{k},$$

with cosine neighbours and self-exclusion [Huh et al., 2024, UniverseTBD et al., 2025]. The primary scale is k=10. All primary results use a frozen train/test split (n_{train}=13,107, n_{test}=3277, seed 0) and a **test-only gallery**: neighbours are sought among held-out objects only (chance $\approx 10/3276 \approx 0.0031$). Direction is Legacy→HSC.

Dense Ridge fits StandardScaler on train X, Y then Ridge with α=1 and an intercept, matching the supervision used for SAE/BSF maps. PCA+Ridge fits independent survey PCAs on an inner training subset (as in our controls), then the same Ridge recipe. Raw-PCA controls instead use a single pooled PCA basis on stacked train embeddings from both surveys—without object pairing—and raw cosine afterwards (Appendix B).

3 Alignment exposes stronger scaling

Mean held-out mKNN@10 is 0.022 for raw Dense, 0.042 for Dense+Ridge, 0.033 for SAE+Ridge, and 0.035 for BSF+Ridge (Figure 1). Dense+Ridge therefore raises recoverable correspondence relative to native geometry. It also steepens family OLS slopes of mKNN versus $\log_{10} P$:

$$\Delta\beta_F = \beta_{\text{Dense+Ridge},F} - \beta_{\text{Dense},F}$$

is positive in all five families (AstroPT $+0.0051$, ConvNeXt $+0.0029$, DINOv2 $+0.0031$, ViT $+0.0103$, I-JEPA $+0.0146$). The primary aggregate is

$$T = \frac{1}{5} \sum_F \Delta\beta_F = 0.00721.$$

Adjacent within-family interactions $D = \Delta M_{\text{Dense+Ridge}} - \Delta M_{\text{Dense}}$ are positive on $9/11$ transitions (mean $+0.0026$).

Three complementary tests support this amplification (Appendix A). A one-sided exact sign test on the five family signs gives $p = (1/2)^5 = 0.03125$. A synchronized shuffle-refit null (B=200) permutes training Legacy↔HSC correspondences once per draw, refits Ridge on every rung, and recomputes T: the null mean is -0.017, none of 200 draws meet or exceed T_{real}, so $p_{\text{perm}} = 1/201 \approx 0.005$. An object bootstrap ($B$=5000) resampling the same held-out IDs across all rungs/methods yields a 95% CI $[0.0034, 0.0111]$ for T; excluding I-JEPA (T_{-I} = mean over four families) gives T_{-I} = 0.00537 with all four remaining families positive. These ask different questions—architecture-level direction, necessity of true pairing for the scaling statistic, and object-level robustness—and are not interchangeable nulls.

Fixed-rank control. To separate representation dimension from alignment capacity, we repeat the comparison at matched PCA rank r=256 with independent survey PCAs (Appendix C). Raw PCA256 native mKNN and PCA256+Ridge use the same 256-dimensional representations; defining $\Delta\beta_{F,256} = \beta_{\text{PCA256+Ridge},F} - \beta_{\text{PCA256},F}$, we obtain $T_{256} = 0.00864$ with $\Delta\beta_{F,256} > 0$ in all five families (sign test p=0.03125; $T_{256}^{(-I)} = 0.00673$). Supervised recoverability therefore steepens with model size even at equal bottleneck dimension.

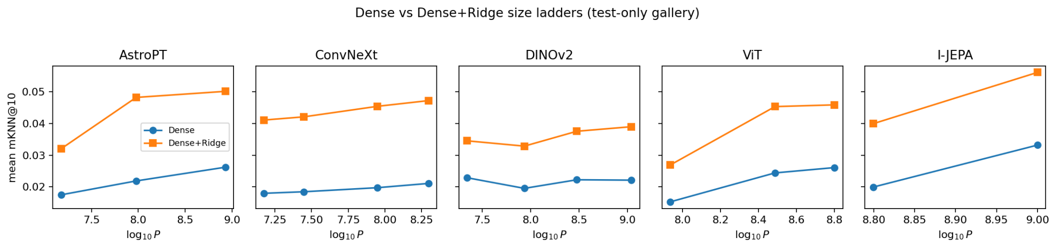

Figure 1: Held-out mKNN@10 versus $\log_{10} P$ (test-only gallery). Dense+Ridge raises the level and steepens the size trend relative to raw Dense in all five architecture families.

Supervised linear alignment therefore amplifies the model-size dependence of recoverable cross-survey correspondence. This is not unsupervised convergence: object pairing is used to fit the map.

4 What kind of alignment?

Beyond similarity transforms. Because cosine neighbourhoods are invariant to a global orthogonal transformation, any increase in mKNN necessarily requires a map that changes native metric geometry. We therefore characterize the learned linear alignment by its effective end-to-end singular spectrum and by empirical local stretch on held-out Legacy edges (Appendix D).

Strong anisotropic deformation. Ridge is fit after train-only $\mathrm{StandardScaler}$ on X, Y; we analyze the composed end-to-end map $A = \mathrm{diag}(\sigma_y)W\mathrm{diag}(1/\sigma_x)$ in original coordinates (W=Ridge `coef_`), not W alone in standardized space. Writing $y \approx Ax + b$ and taking the SVD of A, geometric-mean–normalized singular spectra remain far from flat: $A_{\log} = \mathrm{std}(\log\tilde{\sigma}) \approx 1.76$–$3.15$ (typically ≈ 1.76–1.93), distance from a similarity transform $D_{\mathrm{sim}} \approx 0.89$–$1.00$, and normalized spectral entropy H_{norm} typically ≈ 0.21–0.58 with a few feature-scale–dominated near-degeneracies (H_{norm}=1 is perfectly flat energy). Family slopes of H_{norm} versus $\log_{10} P$ are mixed (2 increasing, 3 decreasing), and recoverability versus H_{norm} has Spearman $\rho \approx 0.04$ across rungs. Alignment becomes more successful with model size without becoming more isotropic (Figure 2b).

Data-supported distortion. On held-out Legacy test edges (k_{edge}=10), local log-stretch $s_{ij} = \log(|A_{\mathrm{eff}}\delta_{ij}|/|\delta_{ij}|)$ has per-rung standard deviation $\sigma_{\mathrm{local}} \approx 0.16$–$0.31$ (Appendix D). A scaled isometry would give constant s_{ij}; the observed spread confirms direction-dependent deformation on the empirical manifold, not only in unused ambient directions. Family slopes of σ_{local} versus $\log_{10} P$ are mixed, so increased recoverability is not accompanied by consistent local metric simplification.

Functionally concentrated maps. Truncating the fitted map A (no refit) shows that roughly 6% of singular directions recover at least 90% of the Dense+Ridge mKNN lift. This concerns the learned map, not the intrinsic rank of the embeddings: most functional contribution to neighbourhood transfer lies in a small leading singular subspace despite global anisotropy.

PCA assists recoverability, not native similarity. Independent-PCA+Ridge improves the *level* of recoverable correspondence at mid/high ranks (mean mKNN $\approx 0.044/0.050/0.055$ at ranks $128/256/512$; 512 on 14/16 rungs with sufficient dimension), rising through the tested ladder rather than peaking at an ultra-low rank (Appendix B). A pooled shared-basis raw-PCA control—fit without pairing—barely changes native mKNN (≈ 0.024 at rank 256 versus 0.022 for Dense), so PCA does not substantially make the surveys more similar in projected native geometry. The PCA×Ridge interaction $I = (M_{\mathrm{PCA+Ridge}} - M_{\mathrm{rawPCA}}) - (M_{\mathrm{Dense+Ridge}} - M_{\mathrm{Dense}})$ is positive at ranks 256 and 512 ($\approx +0.006$ and $\approx +0.010$), indicating that dimensional reduction specifically increases structure recoverable by the supervised map. Relative to Dense+Ridge, PCA does not clearly add a robust further scaling interaction (only 3/5 families positive at $256/512$). The central scaling phenomenon remains Dense+Ridge versus raw Dense.

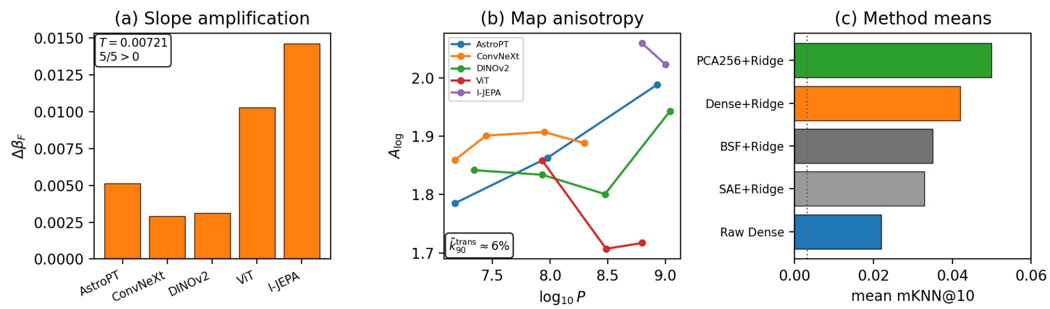

Figure 2: (a) Family slope amplifications $\Delta\beta_F$ (all positive; T=0.00721). (b) Singular-spectrum anisotropy $A_{\log}$ versus model size (maps remain strongly anisotropic). (c) Mean held-out mKNN@10 by method (test-only gallery; dotted: chance).

Sparse representations as negative controls. Earlier SAE/BSF lifts over raw Dense confounded representation change with supervised alignment. With supervision matched, $S_R = M_{R+\mathrm{Ridge}} - M_{\mathrm{Dense+Ridge}}$ averages -0.0085 for SAE (95% family-bootstrap CI $[-0.0120, -0.0043]$; $1/16$ positive) and -0.0062 for BSF (95% CI $[-0.0104, -0.0007]$; $4/16$ positive). Dense+Ridge beats SAE on $15/16$ rungs and BSF on $12/16$. Against Dense+Ridge, SAE slope differences are negative in $4/5$ families and adjacent controlled interactions are mixed ($4/11$ SAE, $6/11$ BSF positive). Sparse SAE/BSF representations do not expose additional cross-survey correspondence beyond an equally supervised dense alignment, and we find no consistent positive sparse-representation$\times$scale interaction beyond that baseline (Figure 2c).

Unpaired nonlinear bottleneck. As a complementary construction that does not use paired cross-survey training objects, a DualEncoder shared latent bottleneck [Jha et al., 2025] recovers above-chance relational correspondence with non-monotonic size dependence (Appendix E). Recoverability therefore depends strongly on the allowed alignment class; the clean size law in these experiments is specific to paired linear recoverability.

5 Discussion and conclusion

The distinction is between native geometric similarity and recoverability under a transformation class. Raw cosine asks whether the two survey representations already induce similar neighbourhood geometry. Ridge asks how much correspondence is accessible after a supervised global linear deformation. The latter exhibits substantially stronger model-size dependence, yet the required maps remain highly anisotropic and do not approach orthogonal or similarity transforms. Scaling therefore increases recoverable common structure without producing evidence that the representations converge toward a shared metric geometry. PCA further increases recoverability despite little effect on native correspondence, suggesting—as a hypothesis rather than a mechanism—that some dense directions interfere with alignment rather than contribute to shared geometry.

Limitations include paired supervision for the main Ridge result, five architecture families (I-JEPA two-rung), a single astronomy cross-survey setting, fixed Ridge α=1 (robustness to other α in Appendix C), and incomplete exploration of local/nonlinear maps and PCA mechanisms. Weak residual locality after global Ridge is measurable but small and is deferred to Appendix A.

In summary: (i) paired linear recoverability scales with model size more clearly than native cosine geometry; (ii) the required transformation remains strongly anisotropic; (iii) compression and sparse representation choices change the level of recoverability but do not overturn the central Dense+Ridge versus Dense scaling law. These results support a weaker notion of Platonic convergence: larger models encode increasingly recoverable shared structure across surveys, even when their native representational geometries remain substantially different.

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499.

Arjun Dey, David J. Schlegel, Dustin Lang, Robert Blum, Kaylan Burleigh, Xiaohui Fan, Joseph R. Findlay, Doug Finkbeiner, David Herrera, Stéphanie Juneau, et al. Overview of the DESI legacy imaging surveys. *The Astronomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

Thomas Fel, Matthew Kowal, Mozes Jacobs, Dron Hazra, Usha Bhalla, Lee Sharkey, Lucius Bushnaq, Satchel Grant, Tal Haklay, Thomas Icard, Can Rager, Michael Pearce, Daniel Wurgaft, Aiden Swann, Fenil Doshi, Siddharth Boppana, Curt Tigges, Nick Cammarata, Thomas Serre, Vasudev Shyam, Owen Lewis, Thomas McGrath, Jack Merullo, Ekdeep Singh Lubana, and Atticus Geiger. Structuring sparsity: Block-sparse featurizers capture visual concept manifolds. *arXiv preprint arXiv:2606.25234*, 2026.

Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025. URL `https://arxiv.org/abs/2406.04093`.

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The Platonic representation hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 20617–20642. PMLR, 2024. URL `https://proceedings.mlr.press/v235/huh24a.html`.

Rishi Jha, Collin Zhang, Vitaly Shmatikov, and John X. Morris. Harnessing the universal geometry of embeddings. In *Advances in Neural Information Processing Systems*, 2025. URL `https://proceedings.neurips.cc/paper_files/paper/2025/hash/4175dee33d6145cb8f0323703d138a53-Abstract-Conference.html`.

Satoshi Miyazaki, Yutaka Komiyama, Satoshi Kawanomoto, Yukiko Fujimori, Takashi Hamana, Hisanori Hayashi, Hiroki Heiguchi, Takashi Hattori, Shiang-Yu Huang, Masatoshi Imanishi, et al. Hyper Suprime-Cam: System design and verification of image quality. *Publications of the Astronomical Society of Japan*, 70 (SP1):S1, 2018. doi: 10.1093/pasj/psx063.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large observation models for astronomy. In *ICML 2024 AI for Science Workshop*, 2024.

UniverseTBD, Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The Platonic universe: Do foundation models see the same sky? *arXiv preprint arXiv:2509.19453*, 2025. doi: 10.48550/arXiv.2509.19453. Accepted to the NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences.

Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16133–16142, 2023.

A Protocol details and statistics

Ladders. Approximate parameter counts (10^6): AstroPTv2 $15/95/850$; ConvNeXtv2 nano/tiny/base/large $15/28/89/198$; DINOv2 small/base/large/giant $22/86/300/1100$; ViT base/large/huge $86/307/632$; I-JEPA huge/giant $630/1000$.

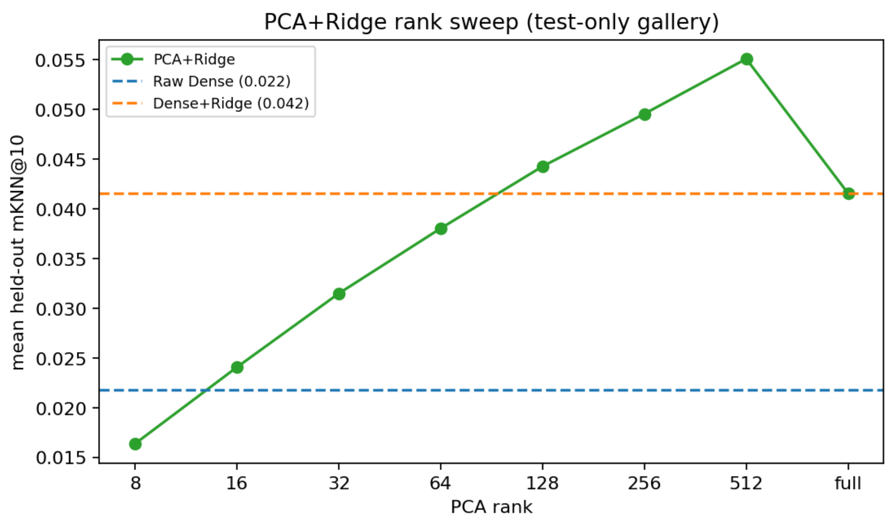

Figure 3: Independent-PCA+Ridge mean mKNN versus rank (test-only gallery), with horizontal references for raw Dense and Dense+Ridge. Performance rises through mid/high ranks; full Dense+Ridge underperforms PCA128–512.

SAE/BSF. SAE width F=2048, seed 0, TopK from available checkpoints ($k \in \{18, \ldots, 23\}$), train-only Ridge and IDF, fixed Legacy→HSC side. BSF uses Vanilla signed codes with the same shared-basis Ridge protocol.

Complementary tests for T. Family sign test: architecture-level directional consistency (n=5). Synchronized shuffle-refit: one permutation seed defines a coherent shuffled training correspondence applied across all sixteen rungs; Dense slopes stay fixed while shuffled-Ridge slopes enter T_b. Object bootstrap: resample test IDs with replacement, recompute mean mKNN per rung/method, refit family slopes, and recompute T.

Residual locality. Global Ridge residuals on test objects are locally structured in source space ($k \in \{10, 25, 50\}$): all 16×3 residual-permutation tests give p=1/201, but absolute residual cosine similarity is ≈ 0.01 and trends versus size are mostly flat. A weak location-dependent correction remains; we do not fit local maps here.

B PCA and raw-PCA controls

Independent-PCA+Ridge means (test-only): ranks 8–64 mostly below Dense+Ridge; rank 128 ≈ 0.044 (13/16 above Dense+Ridge); rank 256 ≈ 0.050 (16/16); rank 512 ≈ 0.055 (14/14 valid). Pooled raw PCA at rank 256: mean ≈ 0.024 versus Dense 0.022. PCA×Ridge interaction means: $\approx +0.006$ at 256 (14/16 positive) and $\approx +0.010$ at 512 (14/14). Raw PCA does not steepen native family slopes relative to Dense.

C Fixed-rank scaling and Ridge-α robustness

At PCA rank 256 with independent survey bases, $T_{256} = 0.00864$ and all five families have $\Delta\beta_{F,256} = \beta_{\mathrm{PCA256+Ridge},F} - \beta_{\mathrm{PCA256},F} > 0$ (sign test p=0.03125). Excluding I-JEPA, $T_{256}^{(-I)} = 0.00673$ with four/four positive. At rank 128, $T_{128} = 0.00227$ with three/five positive (weaker, as expected at a tighter bottleneck).

Ridge regularization sweep ($\alpha \in \{0.01, 0.1, 1, 10, 100\}$, no tuning on test): $T(\alpha) \in [0.0072, 0.0085]$ with all five families positive at every tested α.

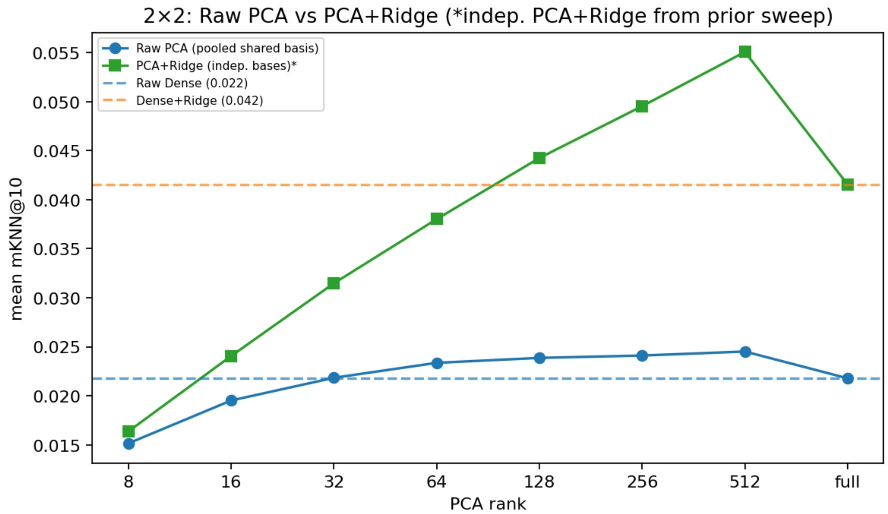

Figure 4: Pooled shared-basis raw PCA versus independent-PCA+Ridge. Raw PCA stays near Dense; almost all of the PCA headline requires Ridge.

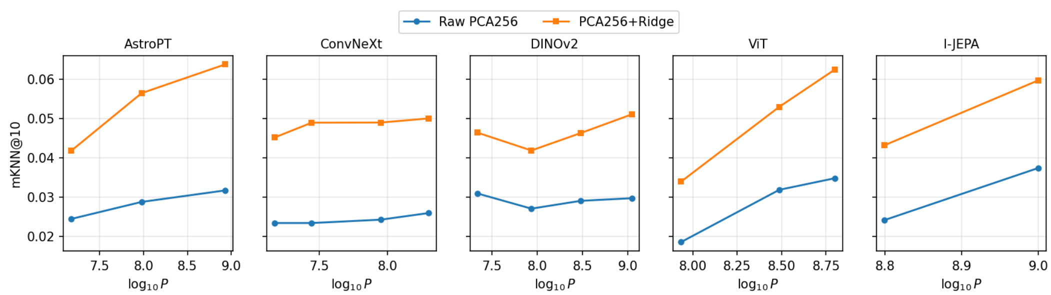

Figure 5: Fixed rank 256: raw independent-PCA mKNN versus PCA256+Ridge by family (test-only gallery). Ridge steepens the size dependence at matched representation dimension.

D Map geometry definitions and local distortion

Effective map and spectral metrics. After train-only $\mathrm{StandardScaler}$, Ridge yields W; the analyzed end-to-end map is $A_{\text{eff}} = \operatorname{diag}(\sigma_y) W \operatorname{diag}(1/\sigma_x)$. Let $\tilde{\sigma}_i$ be singular values normalized by their geometric mean. We report $A_{\log} = \operatorname{std}(\log \tilde{\sigma}_i)$, $D_{\text{sim}} = \min_{c, Q^\top Q = I} |A - cQ|_F / |A|_F$ (optimal $Q^\star = UV^\top$ for $A = U\Sigma V^\top$), and normalized spectral entropy $H_{\text{norm}} = -\sum_i p_i \log p_i / \log d$ with $p_i = \sigma_i^2 / \sum_j \sigma_j^2$.

Local stretch on held-out edges. Within the Legacy test gallery, each point's k_{edge}-NN edges give $\delta_{ij} = x_j - x_i$ and $s_{ij} = \log(|A_{\text{eff}} \delta_{ij}| / |\delta_{ij}|)$. We summarize $\sigma_{\text{local}} = \operatorname{std}(s_{ij})$ and robust spread $R_{\text{local}} = Q_{0.90} - Q_{0.10}$.

E Unpaired relational alignment

Each survey is mapped through a model-specific MLP into a shared latent $Z{=}256$ and decoded back, trained without paired cross-survey objects (reconstruction, RBF-MMD, cycle, and Gram terms; Appendix protocol as previously reported). Held-out relational mKNN@10 lies in 0.012–0.023 versus chance ≈ 0.0035, with non-monotonic size dependence on available ConvNeXt and DINOv2 ladders ($2/6$ adjacent steps positive; Figure 7).

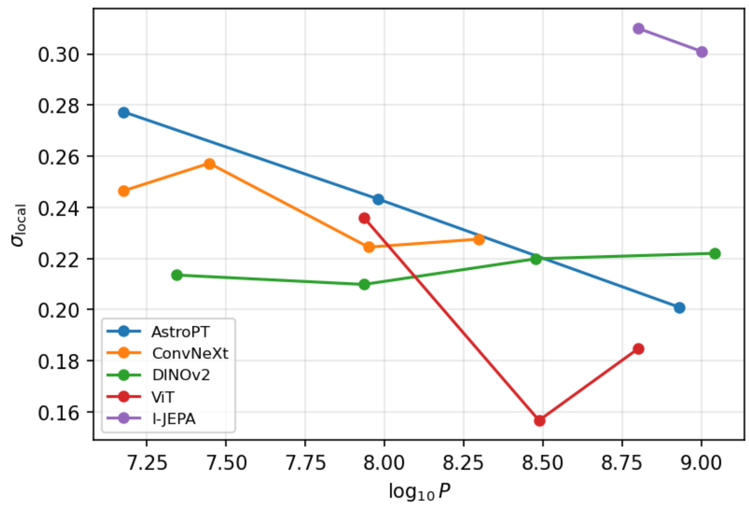

Figure 6: Data-supported local distortion σ_{local} versus $\log_{10} P$ (k_{edge}=10).

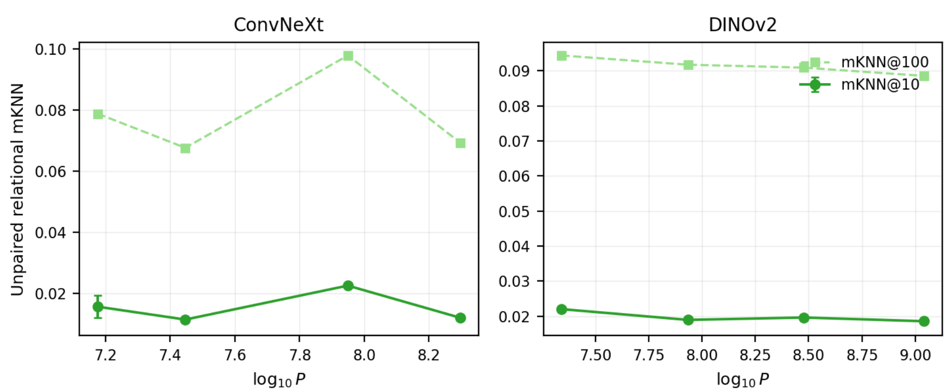

Figure 7: Unpaired DualEncoder relational mKNN versus $\log_{10} P$ (mean $\pm$ seed sd). Curves are non-monotonic across both available families.